# Ideation Phase

## Brainstorm & Idea Prioritization Template

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| **Date** | 22 July 2026 |
| **Team ID** | |
| **Project Name** | India's Agriculture Crop Production Analysis |
| **Maximum Marks** | 4 Marks |

**Brainstorm & Idea Prioritization Template:**

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

This session was used to explore data-driven approaches for the "India's Agriculture Crop Production Analysis" project, so the team could unleash its imagination and start shaping the analysis concept collaboratively.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

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## Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team gathered virtually, reviewed background material on Indian agriculture statistics (state-wise yield, rainfall, and market data), and agreed on the focus problem statement below.

**PROBLEM**

How might we analyze India's crop production data to identify key factors driving yield variation and recommend data-backed improvements for farmers and policymakers?

### *Key rules of brainstorming*

Stay on topic.

Encourage wild ideas.

Defer judgment.

Listen to others.

Go for volume.

Be visual with data.

## Step-2: Brainstorm, Idea Listing and Grouping

Each team member contributed ideas addressing the problem statement. Similar ideas were then clustered into themes.

### *Data Sources & Technology*

- Use state/district-wise crop yield data (Ministry of Agriculture, data.gov.in)
- Integrate satellite / remote-sensing imagery (NDVI) for crop health monitoring
- Build a dashboard visualizing yield trends over the last 10 years
- Use IMD rainfall data to correlate monsoon patterns with yield

### *Environmental Factors*

- Analyze soil type and fertility index by region
- Study irrigation coverage vs rain-fed area yield gap
- Assess impact of climate change / erratic rainfall on major crops
- Compare crop rotation practices across states

### *Economic & Policy Factors*

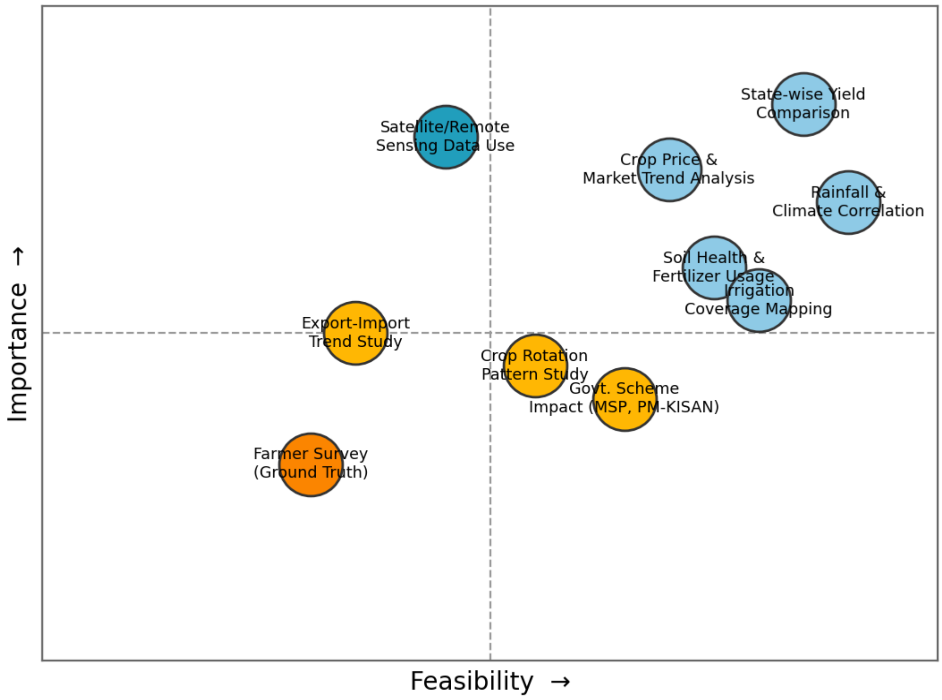
- Evaluate effect of MSP (Minimum Support Price) on crop choice
- Study impact of PM-KISAN and other subsidy schemes
- Track export-import trends for key crops (rice, wheat, cotton)
- Analyze market price fluctuation vs production volume

### *Ground-level Insights*

- Conduct farmer surveys for qualitative insights
- Study adoption rate of modern farming equipment/tech
- Identify high-performing districts as best-practice case studies

## Step-3: Idea Prioritization

The team placed each idea on an Importance vs. Feasibility grid to decide which to pursue first in the analysis project.




**Top priority (high importance, high feasibility):** State-wise yield comparison, rainfall & climate correlation, and crop price/market trend analysis were selected as the starting point for the analysis, since public datasets are readily available and these factors have the most direct effect on production outcomes.

**Next phase:** Government scheme impact and irrigation coverage mapping will be explored once the core yield-trend analysis is complete. Farmer surveys were noted as valuable but lower feasibility given time constraints.